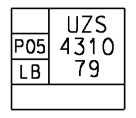
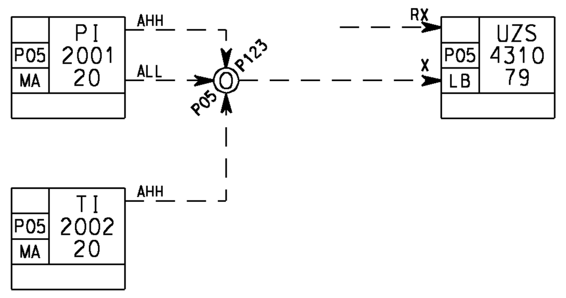
PROCESS SHUTDOWN (PSD)

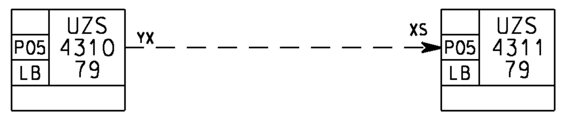


One LB function block shall be used per shutdown level.
The sequence number (four digits) encodes the PSD level:
The first two digits specify the process system, and
the last two digits specify the position of the PSD level
in the PSD hierarchy, that is whether it is higher or lower than
other PSD levels within the same system.

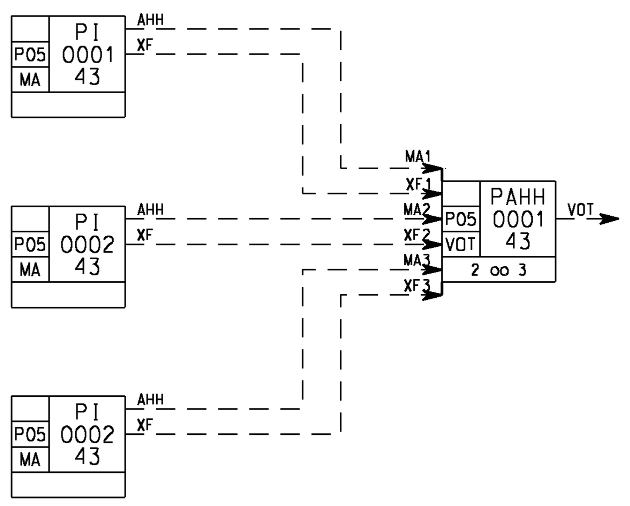
The "PSD hierarchy" is contained in the two drawings:
PSD LOGIC HIERARCHY, C025-W-H000-J-*E-021-01
PSD LOGIC HIERARCHY, WELLHEAD, C025-W-H000-J-*E-022-01



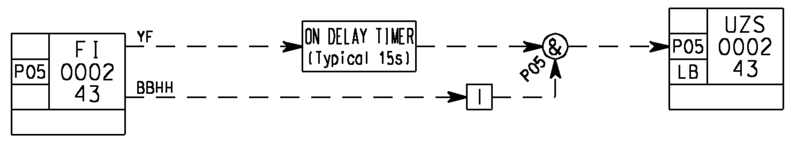
PSD-transmitters with inputs to a PSD-level causing a shutdown.
Inputs directly from transmitters will latch the PSD-level,
i.e. the PSD-level itself has to be reset from the operator station,
eventually with the aid of the RX input terminal at the LB block.



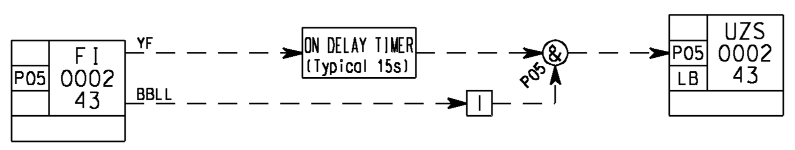
PSD-level tripped from another PSD-level.
The input from another PSD-level does not latch.
I.e. PSD-level 4311 does not need to be reset
after having been tripped from PSD-level 4310.



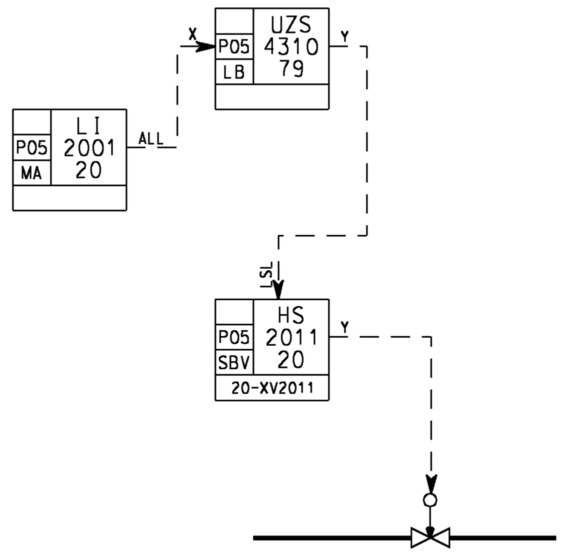
PSD transmitters with shutdown via a voting block
Two out of three devices must trip to perform the action



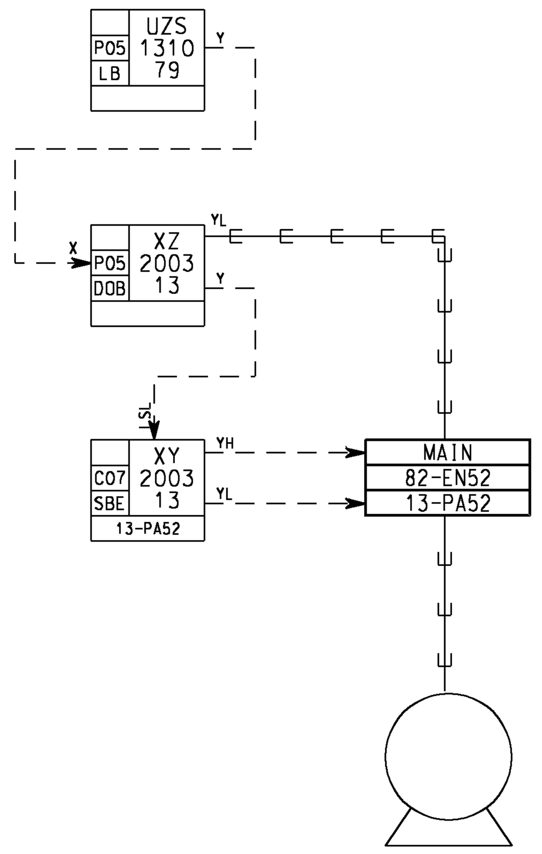
Field transmitter connected to PSD sensing HH value
with time-limited operator possibility
to initiate a PSD
upon transmitter failure



Field transmitter connected to PSD sensing LL value
with time-limited operator possibility
to initiate a PSD
upon transmitter failure

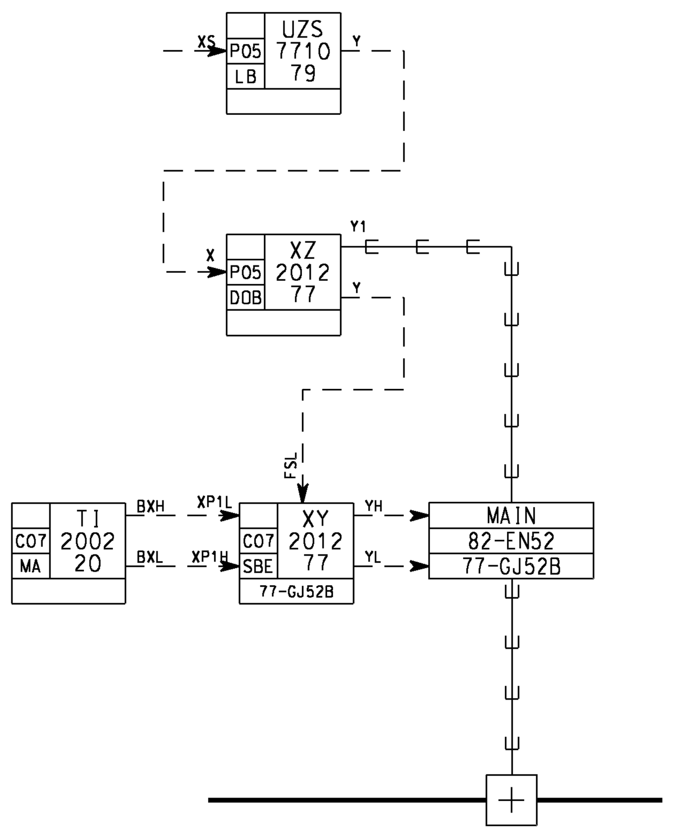


PSD-transmitter which may shut down a PSD-valve.
The valve is latched, i.e. if the level returns to normal,
the valve remains closed. Both the LB and the SBV
must be reset from OS.



A pump is controlled by PCS and can be shut down
from PSD. A trip signal from PSD is going both hardwired
to MCC and by bus to PCS (LSL of SBE). This will aditionally stop
the motor from PCS and avoid fault message from PCS.
When the PSD-level is cleared and reset, the PSD-trip
is automatically reset as the DOB block does not latch.
But the motor is not started because the LSL-shutdown
of the SBE leaves the motor in manual and stopped.

NOTE. The hardwired PSD-output for this trip is tagged 13-XY2003.



A heater is controlled by PCS and may be shut down from a PSD-level.
A shut down signal from PSD is going both hardwired to MCC
(with signal inversion for failsafe PSD output) and
by bus to PCS (FSL of SBE). This will trip the
heater from PCS also.

This shutdown has AUTO RESET, because the DOB block
does not latch, and the SBE is shutdown by means of FSL,
i.e. the heater will continue to respond to the signals
from the temperature measurement after the level measurement
has returned to normal, without need for operator reset.
The heater (i.e. the SBE) must be in auto mode beforehand.

NOTE. The hardwired PSD-output for this trip is tagged 77-XY2012.

Rev.	Date	Description	Made by	Disc. check	Disc. appr.	Contr. appr.	Comp. appr.
A1	30.09.01	AS BUILT	EiL				GS
0	31.03.01	ISSUED FOR CONSTRUCTION	SNo				
03D	10.03.00	RE-ISSUED FOR DESIGN	DD				
02D	26.11.99	ISSUED FOR DESIGN	SNo				
01R	25.10.99	ISSUED FOR REVIEW	SNo				

Acceptance Code: 1☐ 1B☐ 2A☐ 2☐ 3☐ 4☐ Date: Sign:

STATOIL
Kværner Oil & Gas

Project: HULDRA EPC WHPT

Drawing title: SYSTEM CONTROL DIAGRAM

LEGEND 3, PROCESS SHUTDOWN

Tag no.	N/A	
SDL Code		
Scale	NTS	
P.O.No: 480018-P01-01	Projection	A1 (Dwg.Format)
Siemens Doc.No:	Area Code	00 (Syst.Code)
Drawing No: C025-W-H000-J-*E-012-01	1of1 (Sheet)	A1 (Rev.)

Project install | Origin. | Disc./Doc.Type | Seq.No.